

SQL Data Analysis Report

DecodeLabs Industrial Training Kit | Data Analytics – Project 3 | Prepared by Pritesh Raj

Project Overview

This project demonstrates SQL Data Analysis on 1,200 e-commerce orders using an SQLite in-memory database. All queries are written in standard SQL and cover SELECT, WHERE, ORDER BY, GROUP BY, HAVING, COUNT, SUM, and AVG — extracting actionable business intelligence from raw data.

Database Schema: orders table

Column	Data Type	Description
OrderID	TEXT	Unique order identifier (ORD#####)
Date	TEXT	Order date (YYYY-MM-DD)
CustomerID	TEXT	Unique customer identifier
Product	TEXT	Product category (Chair, Laptop, etc.)
Quantity	INTEGER	Units ordered (1–5)
UnitPrice	REAL	Price per unit (■)
ShippingAddress	TEXT	Delivery address
PaymentMethod	TEXT	Payment type (Cash, Credit Card, etc.)
OrderStatus	TEXT	Status: Delivered/Shipped/Pending/Cancelled/Returned
TrackingNumber	TEXT	Shipment tracking number
ItemsInCart	INTEGER	Items browsed before purchase
CouponCode	TEXT	Discount coupon applied
ReferralSource	TEXT	Marketing channel (Instagram, Google, etc.)
TotalPrice	REAL	Total order value = Quantity × UnitPrice

Query 1: Basic SELECT – View Sample Orders

```
SELECT OrderID, Date, Product, Quantity, UnitPrice, TotalPrice, OrderStatus FROM orders LIMIT 5;
```

OrderID	Date	Product	Quantity	UnitPrice	TotalPrice	OrderStatus
ORD200000	2023-01-04	Monitor	5	570.62	2853.1	Shipped
ORD200001	2024-08-23	Phone	2	151.35	302.7	Shipped
ORD200002	2024-02-27	Tablet	5	550.68	2753.4	Cancelled
ORD200003	2023-10-15	Chair	1	273.19	273.19	Returned
ORD200004	2025-05-08	Printer	4	626.01	2504.04	Delivered

■ Insight: Quick snapshot of the data structure. Each order has a unique ID, date, product, and financial details.

Query 2: WHERE – Top Delivered Orders by Value

```
SELECT OrderID, Date, Product, TotalPrice, OrderStatus FROM orders WHERE OrderStatus = 'Delivered' ORDER BY TotalPrice DESC LIMIT 8;
```

OrderID	Date	Product	TotalPrice	OrderStatus
ORD200789	2023-08-17	Tablet	3456.4	Delivered
ORD200632	2023-05-02	Laptop	3390.8	Delivered
ORD201065	2023-10-30	Printer	3334.0	Delivered
ORD200361	2024-06-29	Printer	3299.25	Delivered
ORD200837	2025-06-14	Chair	3277.75	Delivered
ORD201079	2024-06-20	Phone	3170.0	Delivered
ORD200511	2024-03-15	Monitor	2876.2	Delivered
ORD200578	2024-10-10	Monitor	2830.35	Delivered

■ Insight: The highest-value delivered order was ■3,456.40 (Tablet). WHERE filters row-by-row before any aggregation.

Query 3: WHERE (Comparison) – High-Value Orders > ■2,000

```
SELECT OrderID, Product, Quantity, TotalPrice, PaymentMethod FROM orders WHERE TotalPrice > 2000 ORDER BY TotalPrice DESC LIMIT 8;
```

OrderID	Product	Quantity	TotalPrice	PaymentMethod
ORD200789	Tablet	5	3456.4	Online
ORD201122	Monitor	5	3390.95	Online
ORD200632	Laptop	5	3390.8	Gift Card
ORD200469	Chair	5	3384.9	Cash
ORD200328	Tablet	5	3370.2	Online
ORD200107	Printer	5	3353.75	Gift Card
ORD200326	Laptop	5	3352.4	Gift Card
ORD201065	Printer	5	3334.0	Debit Card

■ Insight: 180 orders exceed ■2,000. All are Qty=5 × high UnitPrice — these are bulk/VIP purchases.

Query 4: COUNT + GROUP BY – Orders by Status

```
SELECT OrderStatus, COUNT(*) AS Total_Orders FROM orders GROUP BY OrderStatus ORDER BY Total_Orders DESC;
```

OrderStatus	Total_Orders
Cancelled	250
Returned	247
Pending	237
Shipped	235
Delivered	231

■ Insight: ■ CRITICAL: Cancelled (250) + Returned (247) = 497 orders = 41.4% of all orders. Severe revenue risk.

Query 5: SUM + AVG + GROUP BY – Revenue by Product

```
SELECT Product, COUNT(*) AS Orders, ROUND(SUM(TotalPrice),2) AS Revenue,
ROUND(AVG(TotalPrice),2) AS Avg_Value FROM orders GROUP BY Product ORDER BY Revenue DESC;
```

Product	Orders	Revenue	Avg_Value
Chair	178	195620.11	1098.99
Printer	181	195612.61	1080.73
Laptop	173	192126.56	1110.56
Tablet	179	186568.95	1042.28
Monitor	163	175651.41	1077.62
Desk	170	167459.93	985.06
Phone	156	151722.39	972.58

■ Insight: Chair (■195,620) and Printer (■195,613) lead revenue. Laptop has the highest avg order value (■1,111).

Query 6: AVG + GROUP BY – Revenue by Payment Method

```
SELECT PaymentMethod, COUNT(*) AS Orders, ROUND(AVG(TotalPrice),2) AS Avg_Value,
ROUND(SUM(TotalPrice),2) AS Revenue FROM orders GROUP BY PaymentMethod ORDER BY Avg_Value DESC;
```

PaymentMethod	Orders	Avg_Value	Revenue
Credit Card	234	1127.55	263847.63
Gift Card	230	1070.97	246323.92
Cash	246	1056.04	259786.29
Online	258	1017.22	262442.94
Debit Card	232	1001.56	232361.18

■ Insight: Credit Card customers spend the most on average (■1,128/order). Consider CC-exclusive promotions.

Query 7: GROUP BY + ORDER BY – Revenue by Referral Source

```
SELECT ReferralSource, COUNT(*) AS Orders, ROUND(SUM(TotalPrice),2) AS Revenue FROM orders GROUP
BY ReferralSource ORDER BY Revenue DESC;
```

ReferralSource	Orders	Revenue
Instagram	259	275285.45
Email	250	261808.55
Google	241	250441.48
Facebook	228	250410.9
Referral	222	226815.58

■ Insight: Instagram drives the most revenue (■275,285). Referral programs generate the least (■226,816).

Query 8: HAVING – Products with Revenue > ■1,80,000

```
SELECT Product, ROUND(SUM(TotalPrice),2) AS Revenue FROM orders GROUP BY Product HAVING Revenue
> 180000 ORDER BY Revenue DESC;
```

Product	Revenue
Chair	195620.11

Printer	195612.61
Laptop	192126.56
Tablet	186568.95

■ Insight: HAVING filters AFTER grouping (unlike WHERE which filters rows). 4 of 7 products exceed ■1,80,000.

Query 9: WHERE + AND – Cancelled Laptop Orders

```
SELECT OrderID, Date, Product, TotalPrice, OrderStatus FROM orders WHERE OrderStatus = 'Cancelled' AND Product = 'Laptop' ORDER BY TotalPrice DESC LIMIT 8;
```

OrderID	Date	Product	TotalPrice	OrderStatus
ORD200764	2024-03-10	Laptop	3137.15	Cancelled
ORD200633	2024-04-10	Laptop	3008.6	Cancelled
ORD200608	2024-07-24	Laptop	2715.8	Cancelled
ORD200340	2023-04-23	Laptop	2606.25	Cancelled
ORD201179	2023-01-08	Laptop	2592.75	Cancelled
ORD200805	2024-10-15	Laptop	2360.05	Cancelled
ORD200965	2025-06-18	Laptop	2313.12	Cancelled
ORD200083	2023-08-22	Laptop	2161.64	Cancelled

■ Insight: WHERE + AND narrows to a specific segment. Cancelled Laptop orders are high-value — investigating these could recover significant revenue.

Query 10: COUNT + SUM – Coupon Code Performance

```
SELECT CouponCode, COUNT(*) AS Used, ROUND(SUM(TotalPrice),2) AS Revenue, ROUND(AVG(TotalPrice),2) AS Avg_Value FROM orders GROUP BY CouponCode ORDER BY Used DESC;
```

CouponCode	Used	Revenue	Avg_Value
FREESHIP	313	335036.99	1070.41
No Coupon	309	322401.41	1043.37
WINTER15	292	302483.54	1035.9
SAVE10	286	304840.02	1065.87

■ Insight: FREESHIP is the most-used coupon (313 times, ■3.35L revenue). Customers without coupons also spend similarly.

Query 11: WHERE + GROUP BY – Monthly Revenue (2024)

```
SELECT SUBSTR(Date,1,7) AS Month, COUNT(*) AS Orders, ROUND(SUM(TotalPrice),2) AS Revenue FROM orders WHERE Date LIKE '2024%' GROUP BY Month ORDER BY Month;
```

Month	Orders	Revenue
2024-01	32	38528.08
2024-02	32	36909.57
2024-03	36	36030.9
2024-04	50	49613.14

2024-05	34	27909.11
2024-06	53	68068.54
2024-07	43	42963.98
2024-08	28	31991.07

■ Insight: June 2024 was the peak month (■68,069, 53 orders). May 2024 was the lowest (■27,909, 34 orders).

Query 12: Executive KPI Dashboard – Overall Business Summary

```
SELECT COUNT(*) AS Total_Orders, ROUND(SUM(TotalPrice),2) AS Total_Revenue,
ROUND(AVG(TotalPrice),2) AS Avg_Order_Value, ROUND(MIN(TotalPrice),2) AS Min_Order,
ROUND(MAX(TotalPrice),2) AS Max_Order, COUNT(DISTINCT CustomerID) AS Unique_Customers FROM
orders;
```

Total_Orders	Total_Revenue	Avg_Order_Value	Min_Order	Max_Order	Unique_Customers
1200.0	1264761.96	1053.97	11.39	3456.4	1189.0

■ Insight: Total revenue = ■12,64,762 across 1,200 orders from 1,189 unique customers. Avg order = ■1,054.

SQL Concepts Demonstrated

Query	SQL Concept Used	Business Question Answered
Q1	SELECT, FROM, LIMIT	What does the raw data look like?
Q2	WHERE (equality), ORDER BY	Which delivered orders had highest value?
Q3	WHERE (comparison >)	Which orders exceeded ■2,000?
Q4	COUNT, GROUP BY	How many orders per status?
Q5	SUM, AVG, GROUP BY	Which product generates most revenue?
Q6	AVG, SUM, GROUP BY	Which payment method has highest avg spend?
Q7	GROUP BY, ORDER BY	Which marketing channel drives most revenue?
Q8	HAVING	Which products cross ■1.8L revenue?
Q9	WHERE, AND	How many Laptops were cancelled?
Q10	COUNT, SUM, GROUP BY	Which coupon is most effective?
Q11	WHERE LIKE, GROUP BY	What is month-wise revenue for 2024?
Q12	COUNT, SUM, AVG, MIN, MAX	What are the overall business KPIs?

Tools Used: Python 3, SQLite3, Pandas, ReportLab